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B.Tech. Degree III Semester Examination November 2014

ME 1305 METALLURGY AND MATERIAL SCIENCE (2012 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART A (Answer *ALL* questions)

(8 × 5 = 40)

- I. (a) Draw the miller indices for the following planes.
- (i) (100)
 - (ii) (111)
 - (iii) (101)
 - (iv) (110)
- (b) State Fick's law of diffusion.
- (c) What are the limitations of phase diagram?
- (d) What is precipitative hardening?
- (e) Draw the characteristics for elastic, anelastic and visco-elastic behaviour of materials.
- (f) What is fatigue?
- (g) What are the applications of magnesium alloys?
- (h) Write a note on nanomaterial.

PART B

(4 × 15 = 60)

- II. (a) Find out the co-ordination number, APF of BCC, FCC and HCP crystals along with neat sketches. (10)
- (b) Write a note on Polymorphism. (5)
- OR**
- III. Explain the mechanism of nucleation. (15)
- IV. Explain the Cd-Bi and Pb-Sn phase diagrams. (15)
- OR**
- V. (a) Explain the process of stress relieving of steels. (10)
- (b) What is induction hardening? (5)
- VI. (a) Explain the mechanism of twinning. (7)
- (b) What is recovery, recrystallisation and grain growth? (8)
- OR**
- VII. (a) What is creep and describe its mechanism? (7)
- (b) Explain fracture mechanism. (8)
- VIII. (a) What are the applications of tool steels? (7)
- (b) What are the various types of cast irons? (8)
- OR**
- IX. Describe the role of refractory materials with examples. (15)